BDA501 - BIG DATA ANALYTICS

Individual Assignment 2

Distributed BigEarthNet Processing with Apache Spark

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Abstract

This report presents a distributed BigEarthNet-S2 processing pipeline implemented with Apache Spark. The main experiment uses four selected HDF5 shards containing 52,607 patches from Austria, Finland, Ireland and Lithuania while preserving the official train/validation/test split. Spark executors compute eleven spectral summary features from B02, B03, B04 and B08, Spark SQL produces ten analytical outputs, and a three-class task compares a majority baseline, a training-derived rule, and nearest-centroid models. A 4-versus-8 partition benchmark reduced measured feature-and-write runtime by 18.9% with identical outputs. On the official test subset, the rule reached 0.817 accuracy and 0.858 macro-F1. The implementation is complete for the HDF5-direct path; the main deviation is that original BigEarthNet TIFF files were not packed at full scale in a running HDFS cluster.

1. Dataset scope and processing design

The assignment requires at least 50,000 patches from at least three countries. The submitted run processes 52,607 patches from four countries and keeps the official split unchanged.

Item	Value
Total patches	52,607
Countries	4
Train / validation / test	25,291 / 13,884 / 13,432
Observed labels	18
Main source format	HDF5

The selected shards are bigearthnet_train_p0.hdf5, bigearthnet_train_p1.hdf5, bigearthnet_val_p0.hdf5, and bigearthnet_test_p0.hdf5. The full dataset was not downloaded.



Figure 1. Implemented HDF5-direct processing path; image decoding and feature computation remain on Spark executors.

2. Environment and Step 1 storage

The verified execution uses local Spark/PySpark on Windows. Outputs follow the required BigEarthNet directory convention under a local hdfs_mirror/bigearthnet tree. The mirror keeps the project structure reproducible, but it is not evidence of a running HDFS service. The hdfs and hadoop command-line tools were unavailable; HDFS block size, replication factor and live DataNode count are therefore not inferred.

Environment item	Verified value
Execution mode	Local Spark / PySpark on Windows
HDFS CLI	Not available
HDFS block size / replication	Not available
Local results root	hdfs_mirror/bigearthnet
Spark UI during benchmark	http://127.0.0.1:4040

Storage evidence is provided in three forms: (1) the main HDF5-direct path; (2) a measured TIFF-to-Parquet pilot for the small-files mechanism; and (3) a full 52,607-row HDF5-derived raw-band Parquet pack containing B02/B03/B04/B08 byte columns.

Storage experiment	Rows/patches	Files	Size
Raw TIFF pilot	24	96	2.66 MiB
Packed Parquet pilot	24	8	2.06 MiB
Full HDF5-derived raw-band Parquet	52,607	136	6.05 GB (5.63 GiB)

The pilot reduced file count from 96 to 8 and size from 2.66 MiB to 2.06 MiB in 26.147 s. The full raw-band pack demonstrates large-scale Parquet packing from HDF5-derived bytes, but it is not presented as full original-TIFF packing because the original TIFF corpus was not downloaded.

3. Distributed feature engineering

scripts/hdf5_features.py uses Spark mapPartitions. Each partition opens the required HDF5 shard, reuses the file handle across rows, reads the indexed image, selects B02, B03, B04 and B08, and emits a typed row. The schema includes error_flag and error_message so corrupt input can be recorded instead of terminating the full job.

The eleven numerical features are the mean and standard deviation of each selected band plus NDVI, NDWI and brightness. NDVI is computed from B08 and B04, NDWI from B03 and B08, and brightness is the mean of the four band means.

3.1 Validation and partition performance

Validation check	Result
Input / output rows	52,607 / 52,607
Error rows	0
Invalid NDVI / NDWI rows	0 / 0
Null / NaN / Inf values	0
Independent NumPy checks	3 / 3 exact matches
Original main-run elapsed time	76.136 s

The feature profile also contains Spark describe() statistics (count, mean, stddev, min and max) for all eleven features. Three independent NumPy recomputations match the Spark values exactly, with maximum absolute difference 0.0.

The same 52,607-patch feature job was then benchmarked with 4 and 8 partitions. The measured region covers feature extraction and Parquet writing; validation and cross-run comparison occur afterwards.

Partitions	Input	Output	Errors	Runtime
4	52,607	52,607	0	60.545 s
8	52,607	52,607	0	49.085 s

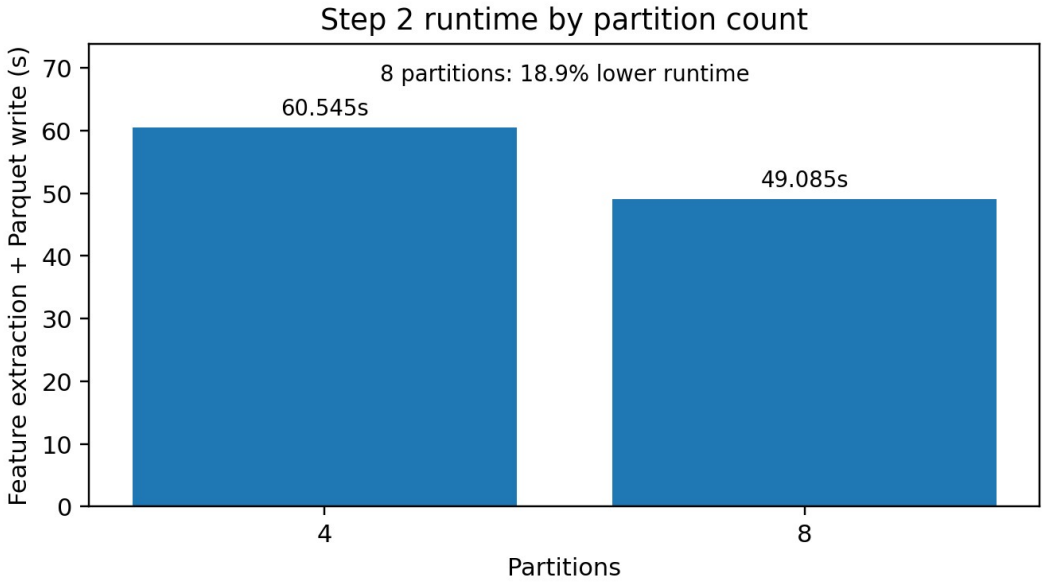


Figure 2. Step 2 benchmark. Eight partitions reduced measured runtime by 18.9% relative to four partitions on this machine.

The two outputs contain the same 52,607 patch IDs and have maximum absolute feature difference 0.0 across all eleven features. This supports a local performance improvement without a numerical change; it is not a general claim of linear scalability. Spark statusTracker data captured job/stage/task execution. A literal browser screenshot of the Spark UI was not preserved.

4. Spark SQL analysis

The feature table is registered for Spark SQL analysis and result tables are saved under the project results tree. Ten outputs cover class distribution, label cardinality, country/split composition, co-occurrence, feature summaries and two custom questions.

Analysis	Purpose
Class distribution + by split	Exploded multi-label frequency and split comparison
Labels per patch	Multi-label cardinality
Country + country x split	Geographic coverage and split integrity
Top-10 co-occurring pairs	Frequent label combinations
Mean features by country/class	Spectral summaries
Two custom queries	Single-label rate by country; NDVI/brightness by split

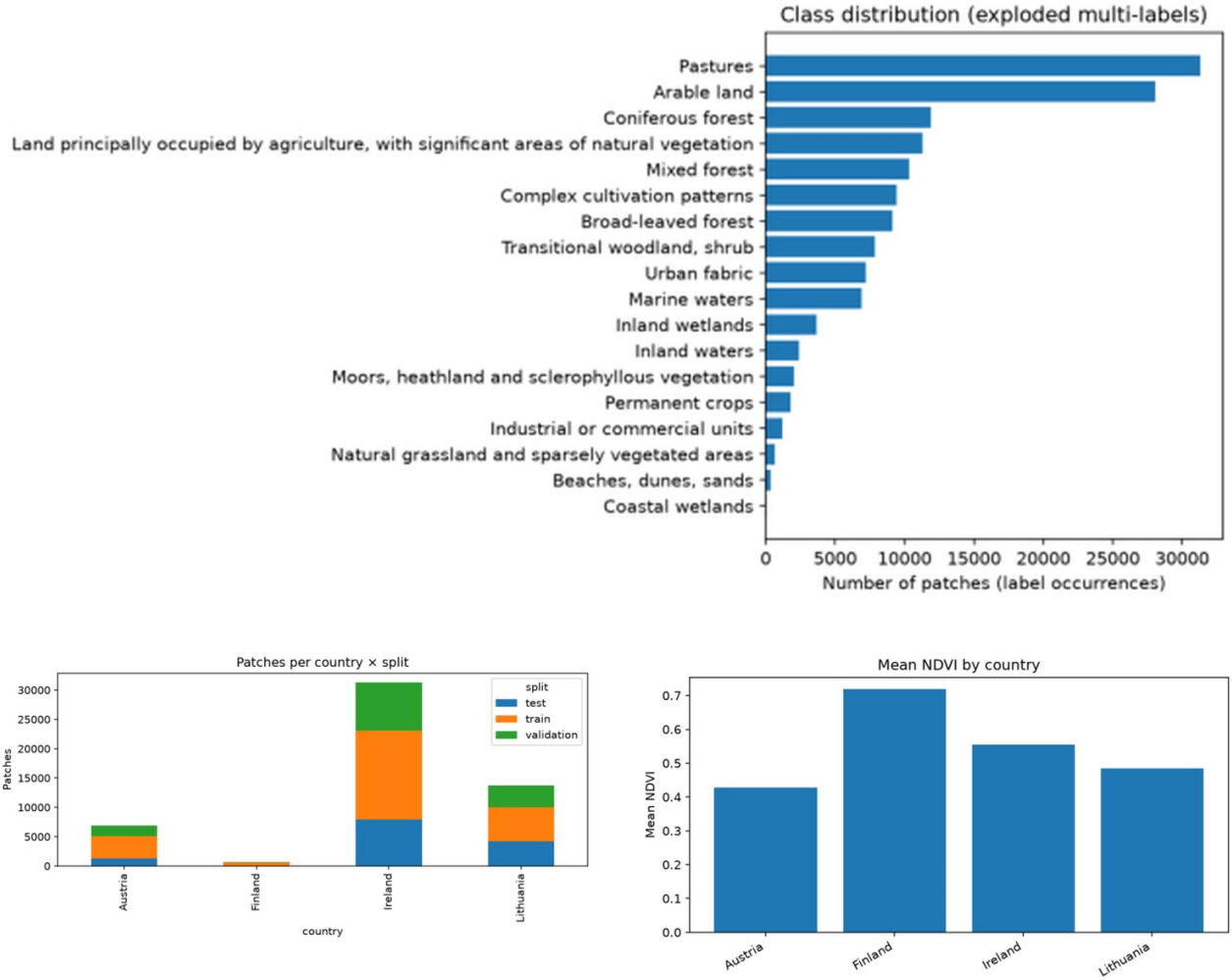


Figure 3. Spark SQL results: exploded class distribution, country x official split counts, and mean NDVI by country.

The class distribution is long-tailed: Pastures and Arable land are the largest labels, while several classes occur far less frequently. Because the dataset is multi-label, exploded label counts exceed the patch count. Country coverage is also uneven, but the country-by-split result confirms that the official train/validation/test grouping is retained. Mean NDVI is highest for Finland (~0.72), followed by Ireland (~0.55), Lithuania (~0.48) and Austria (~0.43); these are descriptive subset statistics, not causal country effects. The imbalance motivates reporting macro-F1 alongside accuracy in the classification task.

5. Three-class classification

The task keeps patches containing exactly one of three selected targets: Pastures, Arable land and Marine waters. The official split is preserved: training data supplies thresholds and standardisation statistics, validation supports reduced-feature selection, and the test split is reserved for final scores.

Split	Train	Validation	Test
Rows	16,835	7,551	6,371

Four methods are evaluated: majority baseline, a training-derived spectral rule, nearest centroid with all eleven features, and nearest centroid with five reduced features. The rule follows the strong ordering of training mean NDVI: Marine waters -0.252, Arable land 0.479, and Pastures 0.725. The reduced centroid uses ndwi, mean_b08, ndvi, brightness and std_b08.

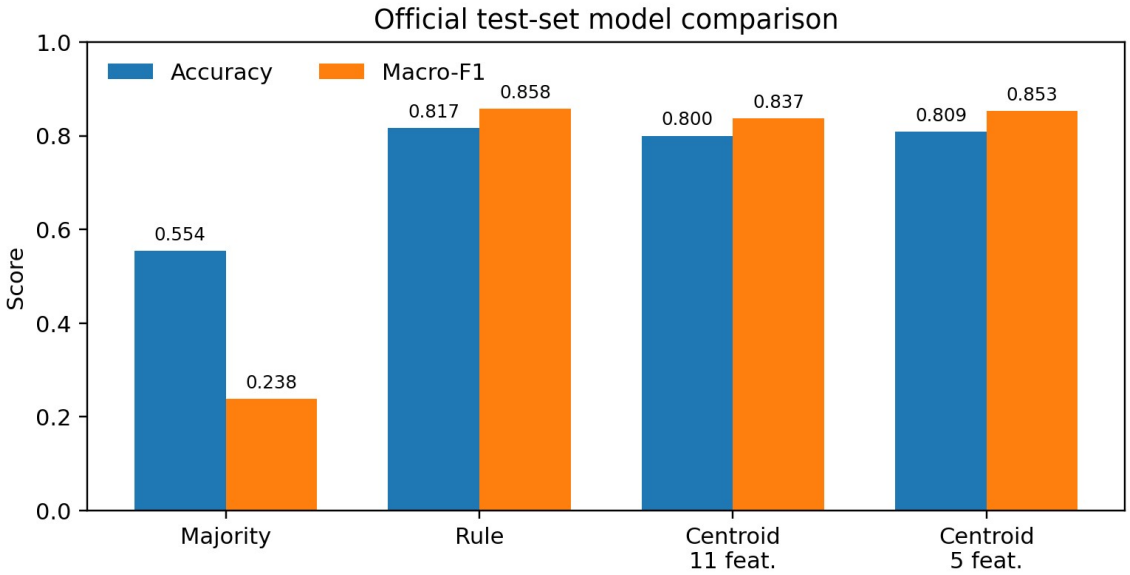


Figure 4. Accuracy and macro-F1 on the official test subset.

Model	Accuracy	Macro-F1
Majority baseline	0.554	0.238
Training-derived rule	0.817	0.858
Centroid - 11 features	0.800	0.837
Centroid - 5 features	0.809	0.853

5.1 Per-class behaviour

Model	Pastures F1	Arable F1	Marine F1
Rule	0.846	0.766	0.963
Centroid - 11	0.835	0.737	0.938
Centroid - 5	0.827	0.780	0.950

Marine waters is consistently easiest because its NDVI and NIR behaviour is well separated from the vegetation classes. The main ambiguity is between Pastures and Arable land. The reduced centroid improves Arable-land F1 and macro-F1 relative to the full centroid, indicating that removing weak or redundant dimensions can help the distance-based classifier. Spark result folders also contain confusion matrices and per-class precision/recall/F1 outputs.

6. Discussion and limitations

The main technical result is a reproducible HDF5-direct Spark pipeline that processes more than 50,000 image patches through executor-side feature extraction, SQL analysis and Spark-based classification. Validation shows one-to-one row preservation, no recorded feature errors, bounded NDVI/NDWI values, no null/NaN/Inf values and exact agreement with three independent NumPy checks. The partition benchmark additionally shows that moving from 4 to 8 partitions improved the measured feature-and-write runtime on this machine without changing the output values.

The strongest test result is the training-derived rule at 0.858 macro-F1. This result is specific to the selected classes, whose spectral behaviour is strongly ordered; it should not be interpreted as evidence that rule-based modelling is generally superior. The centroid models remain useful compact baselines. Their main limitation is representational: patch-level means and standard deviations discard spatial texture, edges, shapes and local arrangements that an image model could exploit.

Two implementation limitations remain. First, the assignment specifies full-scale packing of original per-band TIFF files into HDFS Parquet. This submission instead combines a measured TIFF pilot, a full HDF5-derived raw-band Parquet pack, and an HDF5-direct main pipeline. Second, a running HDFS service was not available in the verified environment, so real block size, replication factor, DataNode count and hdfs dfs command evidence are unavailable. Spark execution evidence was captured through statusTracker, but a browser screenshot of the Spark UI was not preserved.

7. Reproducibility and conclusion

Reproducibility is supported by the project scripts, README.md, requirements.txt, JSON evidence files under reports/, SQL/classification result folders, and benchmark outputs under data/benchmark. Reported counts, timings, validation checks and metrics are taken from those artifacts rather than estimated values.

Overall, the project satisfies the required processing scale and completes the core Spark workflow for 52,607 BigEarthNet patches: distributed feature engineering, validation, SQL analysis, visualisation and three-class classification. The best test result is 0.858 macro-F1 from the training-derived rule; the reduced centroid reaches 0.853. The remaining gap is infrastructure/storage compliance rather than the analytical pipeline itself: original TIFF files were not packed at full scale in a real HDFS deployment.

AI acknowledgement

Generative AI was used to assist with planning, coding support, organisation and language polishing. Dataset counts, timings, feature validation results, SQL outputs and classification metrics reported here are taken from the project's own artifacts and command outputs. No text or results from another student's report were copied into this submission.